Learned variable-support priors accelerate symbolic regression of physical laws

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Abstract

Symbolic regression recovers closed-form, interpretable physical laws from data, but its cost grows steeply with the number of candidate variables and operators, most of which never appear in the governing equation. Here we show that a learned front-end—DenoisedSR—that predicts which variables are relevant (a support prior) improves and accelerates symbolic regression without changing the backend solver. The front-end couples a graph-attention network with a statistical-feature model, and recovers the true variables with recall above 0.99 (perfect recall on 99–100% of tasks) across three physics suites, so it essentially never discards a real variable. Supplied to the evolutionary backend PySR under a fixed time budget, the prior doubles exact-law recovery (from 30% to 60%), shrinks the candidate-column set by 86%, and accelerates time-to-solution by up to 6×; on black-box tabular data it improves accuracy while a same-size random-support control degrades it, isolating learned guidance from dimensionality reduction. The approach is solver-agnostic, adds ∼40 ms per task, and keeps final law selection with a verifiable symbolic backend.

1 Introduction

The automated discovery of governing equations from observation is a long-standing goal of the physical sciences[1–3]. Symbolic regression (SR) addresses it directly: rather than fitting an opaque function, SR searches the space of mathematical expressions for a compact formula that both fits the data and remains human-interpretable[4,5]. The resulting equation can be dimensionally checked, extrapolated and falsified, properties that black-box predictors lack, which has made SR central to data-driven physics, from rediscovering conservation laws to proposing constitutive relations[2,6,7].

The central obstacle is combinatorial. An SR backend searches over expression trees whose leaves are drawn from the input variables and whose internal nodes are drawn from an operator library; the number of candidate trees grows exponentially with the number of admissible leaves and operators and with tree depth[8,9]. In physics this cost is largely wasted: a governing law typically uses only a handful of the measured variables and a small subset of the operator library, yet a solver run over all columns and operators still explores an enormous number of branches that cannot appear in the answer. Established systems attack this from the solver side with dimensional analysis, decomposition, sparsity priors or efficient enumeration[2,6,10,11]. A separate line learns to decode the equation directly from numerical data with transformer or reinforcement-learning models[12–15], sometimes coupling a neural proposer with explicit search[16,17] or multimodal pre-training[18,19], with

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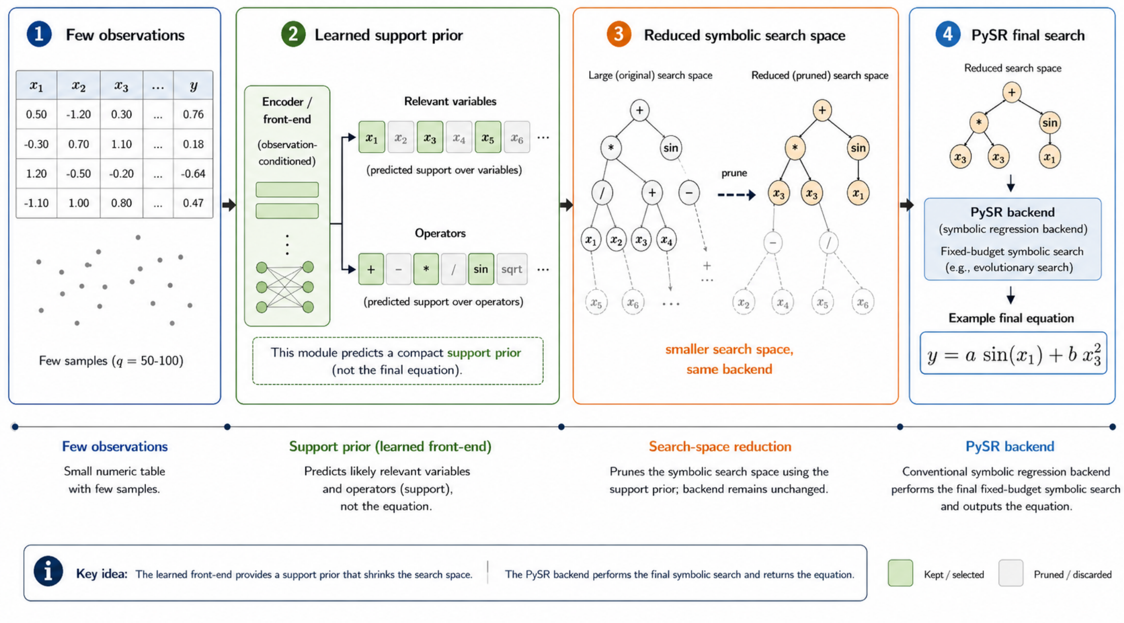

Fig. 1. DenoisedSR narrows the symbolic-regression search space. From a small set of observations the learned front-end predicts a compact support prior (relevant variables and, optionally, operators); a conventional symbolic-regression backend then performs the final search over the reduced space and returns a verifiable equation. The neural model proposes where to look; the backend decides what the law is.

e-graph, diffusion and generative-latent variants[20–22]. These end-to-end models advance quickly but remain unreliable at *assignment*: they often recover the right ingredients yet place them in the wrong terms, so exact symbolic equivalence stays low.

We take a complementary route that sidesteps this brittleness. Instead of designing a faster search or decoding the whole equation, we *narrow the space the search must explore* (Fig. 1). Given a small observation set $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{q}$ with q as small as 50–100, a learned model—DenoisedSR—predicts which input columns (and, as an auxiliary, which operators) are likely to participate in the law, and passes this *support prior* to a conventional SR backend that performs the final symbolic search and model selection. Predicting support is a far easier and more robust target than decoding the full expression, and a high-recall variable filter hands the backend an exponential head start while essentially never making the true law unreachable.

In this work we cast variable- and operator-support prediction as a reusable, backend-agnostic SR front-end; we present a graph-attention support predictor, ensembled with a statistical-feature model, that recovers the true variables with recall above 0.99 across three physics suites; and we show that supplying this prior to a fixed-budget backend doubles exact-law recovery, reduces the search space by 86% and accelerates time-to-solution by 3–6×, while a same-size random-support control confirms that the benefit is learned search guidance rather than dimensionality reduction.

2 Results

DenoisedSR recovers the true physical variables

We first evaluate the support prior intrinsically, without any backend. For each task the front-end receives $q = 100$ noisy observations whose columns include the genuine variables together with

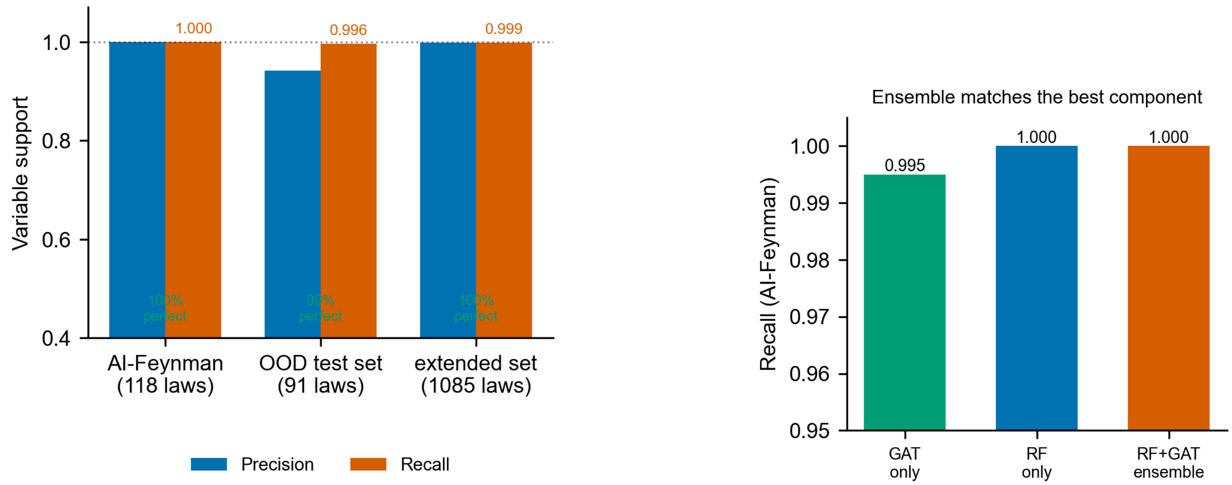

Fig. 2. The front-end recovers essentially every true variable. *Left*: precision and recall of predicted variable support across three physics suites; recall is ≥ 0.996 with perfect recall on 99–100% of tasks. *Right*: ablation on AI-Feynman—the random-forest and graph-attention models each reach recall ≥ 0.995 and the ensemble matches the best component.

random "distractor" columns, and must predict the true variable subset. Across three suites of physical laws—AI-Feynman (118 equations), an out-of-distribution (OOD) test set of 91 curated physical laws, and an extended physics-law set (1085 equations)—the predictor attains recall 1.000, 0.996 and 0.999 respectively, with perfect recall on 99–100% of tasks (Fig. 2). It essentially never drops a variable that the law uses; its only error mode is occasionally retaining a few extra noise columns, which merely costs the backend a little additional search rather than making the law unreachable. Precision stays high (0.94–1.00), reflecting an intentional high-recall operating point: removing a true variable is fatal to recovery, whereas keeping a spurious one is recoverable downstream.

The result is robust and not attributable to a single component. An ablation on AI-Feynman (Fig. 2, right) shows the random-forest and graph-attention models each reach recall ≥ 0.995 and their ensemble matches the best component (1.000); recall stays at 1.000 as the number of distractor columns is varied from 1 to 30. A naive statistical baseline trained on few records with name-matched labels reaches only recall 0.48, confirming that the learned predictors contribute the decisive signal.

A high-recall prior strengthens a fixed-budget solver

We next insert DenoisedSR before PySR[5] and compare three conditions at a fixed 10 s per-task budget on a 30-equation Feynman subset with $\sim$20 distractor columns: (A) full PySR over all columns and the full 17-operator library; (B) the variable prior, restricting the columns to the predicted support; and (C) additionally restricting the operator library. The variable prior raises exact-law recovery (held-out $R^2 \geq 0.999$) from 9/30 (30%) to 18/30 (60%), lifts mean held-out R^2 from 0.855 to 0.967 (averaged over all 30 tasks, with non-finite failures floored at -1), reduces the average column count from 23.2 to 3.2 (an 86% reduction) and halves the effective search vocabulary from 40 to 20 (Fig. 3).

The mechanism is concrete. Over many irrelevant columns, an unguided solver can reach a deceptively high R^2 with a physically meaningless formula built from distractor variables (Table 1); for the Boltzmann relation $S = k_{\mathrm{B}} \ln W$, for example, full search returns $1.31\, e^{\cos(\cdots)}$ at $R^2 = 0.996$.

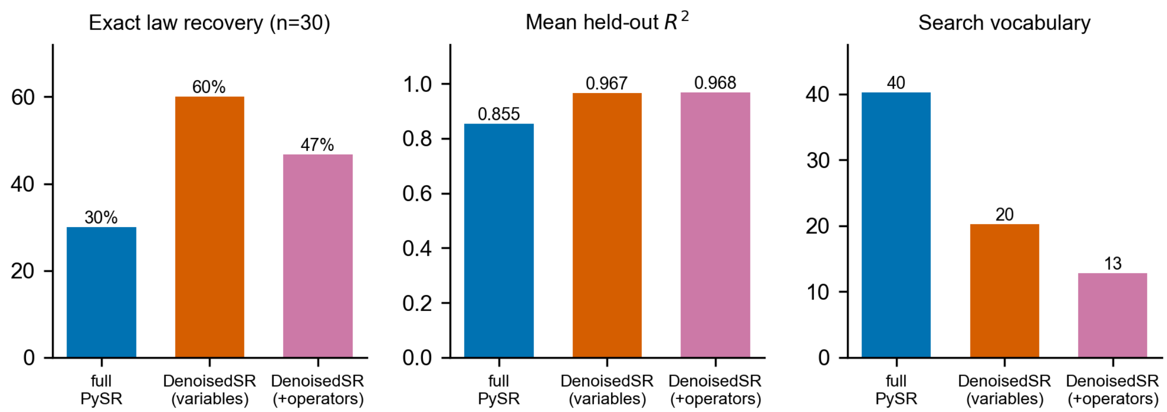

Fig. 3. A variable prior strengthens a fixed-budget solver. Full PySR versus the DenoisedSR variable prior versus the additional operator prior on a 30-equation Feynman subset (10 s budget, $\sim$20 distractors). The variable prior doubles exact recovery (30% to 60%), raises mean held-out R^2 (0.855 to 0.967) and halves the search vocabulary (40 to 20); the operator prior cuts the vocabulary further but does not improve exact recovery.

Table 1. Unguided search overfits to distractor variables. On laws padded with random distractor columns, full PySR can reach a high held-out R^2 with a physically meaningless expression that uses the wrong variables. DenoisedSR predicts the correct variable support (essentially perfect recall, Fig. 2), so these spurious fits are removed from the search space.

Physical law	True form	Full-PySR R^2	Recovers law?
Boltzmann entropy	$S = k_\mathrm{B} \ln W$	0.996	✗ spurious
Chirp mass	$M_c = (m_1 m_2)^{3/5}/(m_1{+}m_2)^{1/5}$	0.9999	✗ spurious
Nernst equation	$E = \frac{RT}{zF} \ln(c_\mathrm{out}/c_\mathrm{in})$	-6.5×10^3	✗ diverged

By removing the distractor columns, DenoisedSR makes such spurious overfits unreachable, so a far larger fraction of budget-limited runs land on the true law.

More data helps DenoisedSR but hurts unguided search

A striking consequence of pruning is how the methods scale with data (Fig. 4). On a 50-equation Feynman set with 10 distractors, increasing the observations from $q = 50$ to $q = 100$ *lowers* the exact-recovery rate of full search (from 16% to 8%): more data slows each backend evaluation and exposes additional spurious high-R^2 fits among the distractor columns. Over the same range DenoisedSR *improves* (from 18% to 24%) and tracks the oracle-support upper bound, because it first removes the irrelevant columns and then lets the backend benefit from the extra data. The front-end thus lets a solver actually exploit more observations instead of drowning in them.

The gain is search guidance, not dimensionality reduction

A natural concern is that any reduction in column count would help. We control for this with a same-size random-support baseline that retains the same number of columns but selects them without using the observations. On 24 PMLB/SRBench black-box tabular tasks (three seeds, $q = 100$, 5 s budget), DenoisedSR raises mean held-out R^2 from 0.347 to 0.495 with 10 distractor variables and from 0.315 to 0.488 with 20 distractors, using only 31% and 18% of the input columns;

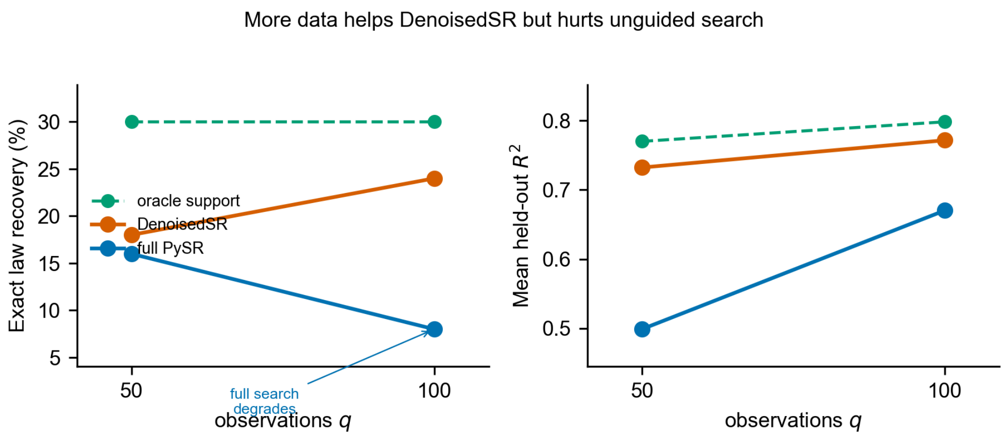

Fig. 4. More data helps DenoisedSR but hurts unguided search. Exact-law recovery (left) and mean held-out R^2 (right) versus the number of observations q, on a 50-equation Feynman set with 10 distractors. Full search degrades with more data while DenoisedSR improves and approaches the oracle-support upper bound.

the same-size random support instead degrades performance to 0.220 and −0.028 (Fig. 5). At matched column count the random control is far worse than full search whereas the learned prior is better—direct evidence that the front-end supplies observation-conditioned information about *which* columns matter, and that it helps even when no symbolic ground truth exists.

Pruning the search space accelerates time-to-solution

Because variable filtering shrinks the terminal set of the expression tree from ∼24 to ∼4 symbols—an exponential reduction in branching—it also shortens the time to reach an exact fit. Measuring time-to-solution with early stopping on exact recovery, DenoisedSR yields speedups of up to 6× on non-trivial physics formulas (Fig. 6); in some cases it converts a within-budget failure of full search into a fast success. The overall median speedup is ∼1.5×, pulled down by deep-tree formulas whose bottleneck is expression depth rather than variable count and which therefore gain little from variable pruning. Front-end inference adds only ∼40 ms per task, near-constant in the number of observations and negligible against the seconds of backend search it saves.

An operator prior helps less than a variable prior

Restricting the operator library in addition to variables (condition C) recovers 14/30 (47%) exact—below the variable prior (60%) but still above full search (30%), with the same mean held-out R^2 as the variable prior (0.968; Fig. 3). Operators are intrinsically harder to infer from small, noisy domains, and a mis-removed operator is fatal, so we report the operator prior as an ablation rather than a primary result. Its most plausible use is as a safeguard: when full-operator search returns a pathological fit built from exotic operators, restricting the library to the predicted operators should steer the search back toward a sane solution; quantifying this rescue at the level of individual recovered expressions requires a backend run that logs the before-and-after formulas, which we leave to future work.

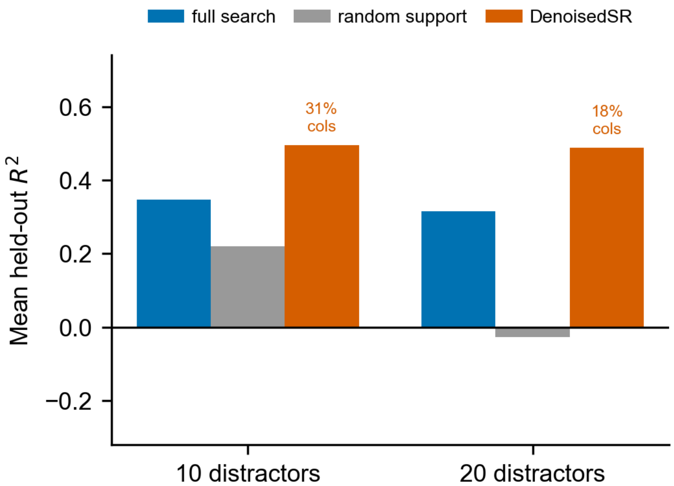

Fig. 5. The gain is search guidance, not fewer columns. Mean held-out R^2 on 24 PMLB/SRBench black-box tasks (three seeds, fixed 5 s budget) at 10 and 20 distractor variables. At matched, reduced column count (labels show columns relative to full search) DenoisedSR improves over full search while a same-size random support degrades far below it.

3 Discussion

These results support a simple thesis: for symbolic regression of physical laws, a learned support prior is a high-leverage, low-risk intervention. Because a high-recall variable filter essentially never makes the true law unreachable, it can be applied unconditionally; because it removes an exponential number of irrelevant branches, it materially improves both the success rate and the speed of a fixed-budget backend. DenoisedSR is solver-agnostic—its output is a reduced variable/operator set that any tree-search, enumeration or generative backend [5,11,22] can consume—and cheap enough to prepend to every run. The same-size random-support control shows the benefit is genuine search guidance rather than dimensionality reduction, complementing the dimensional and physics-aware priors that already strengthen scientific SR [2,7].

Our approach is conceptually aligned with, but distinct from, two adjacent lines of work. Physics-aware neural models embed structure such as Hamiltonian or Lagrangian form, differential-equation structure, or symbolic distillation of trained networks [23–26], and concept-driven or abductive discovery systems [27–29] likewise aim to keep the final law symbolic and verifiable; we contribute the upstream step of deciding *which measured quantities enter the law at all*. Because the front-end consumes *sets* of observations, it builds on permutation-invariant and set-attention architectures [30,31], and its representation could be sharpened with the contrastive and geometric objectives used to prevent representational collapse or encode hierarchy [32–34].

The honest limitation, shared with direct neural SR [13,14], is that the neural component cannot yet reliably *assign* variables and operators to the correct positions in the expression; this is precisely why we leave final term assignment to a symbolic, verifiable backend rather than emitting the equation end-to-end. Closing this assignment gap—through richer compositional latents, explicit constant and exponent recovery, and objectives that keep the observation and formula representations aligned [18,35]—is, in our view, the central open problem in moving from a front-end prior toward reliable end-to-end neural discovery [36–38]. Even short of that goal, DenoisedSR is immediately useful: it makes existing physical-law solvers both more successful and several times faster, with a provably safe failure mode.

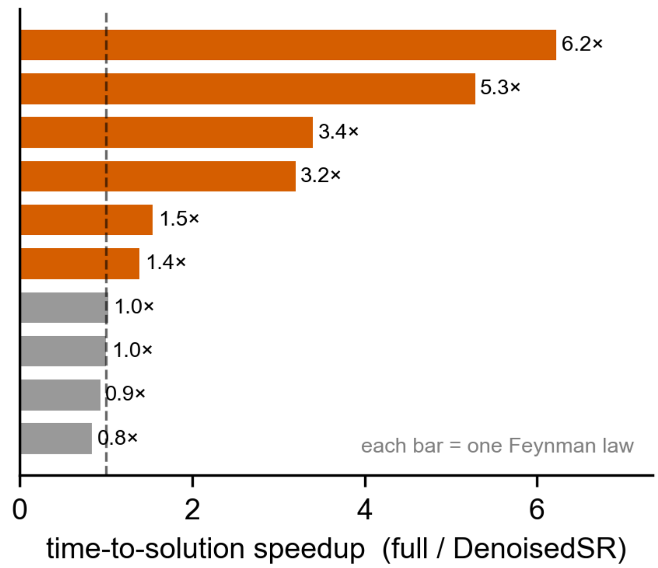

Fig. 6. Pruning accelerates time-to-solution. Per-formula speedup (full-search time divided by DenoisedSR time) with early stopping on exact recovery. Non-trivial physics formulas solve up to 6× faster (median ∼1.5×); deep-tree formulas (depth-, not variable-limited) show little gain.

4 Methods

Problem setup. Each task provides $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{q}$ with $\mathbf{x}_i \in \mathbb{R}^m$, where only an unknown subset $S^\star \subseteq \{1, \dots, m\}$ of columns appears in the target law $f^\star$. DenoisedSR predicts $\hat{S} \approx S^\star$ (and an optional operator set $\hat{O}$) and passes it to an SR backend run under a fixed time budget. Tasks are padded with random distractor columns to emulate the realistic regime in which a measurement table contains many irrelevant channels.

Front-end architecture. The variable-support predictor is an ensemble of two complementary models. (1) A graph-attention network[30,31] operates on a graph whose nodes are the input columns; node features combine per-column statistics (moments, correlation with the target, monotonicity and dimensional descriptors) and edges encode a variable co-occurrence prior learned from a knowledge graph of physical laws together with observation-derived correlation. Attention over this graph lets the model reason about which columns *jointly* support a plausible law rather than scoring columns in isolation. (2) A random forest scores each column from the same per-column features. The final score is $s(\text{col}) = \text{RF}(\text{col}) + \sigma(\text{GAT}(\text{col}))$, and a column is retained if $s \geq \tau$ with a low threshold ($\tau = 0.10$) chosen to favour recall. An analogous graph-attention model with graph-level pooling produces the multi-label operator prediction used for the optional operator prior.

Training. The predictors are trained only on internal physical-law records—formula templates with sampled observation tables drawn from a curated knowledge graph of physical laws—never on the external evaluation suites. Training pools of 8000–40 000 formulas were used, with observation tables pre-sampled across variable ranges and a 10–30 distractor curriculum so the front-end is robust to the number of irrelevant columns at test time. An adversarial variant (a generative-adversarial graph-attention model) was explored but gave no consistent gain and is not used.

Backend integration and evaluation. We use PySR[5] as the downstream backend because it exposes variable and operator restrictions directly and is reproducible locally; the same prior interface applies to enumeration[11] and generative[22] backends. All comparisons fix the backend budget across conditions, so the only changed quantity is the search space exposed to the solver. Exact recovery denotes held-out $R^2 \geq 0.999$. Support quality is measured by precision and recall against the known law variables, at the deployed ensemble threshold $\tau = 0.10$; the same-size random-support control retains the same number of columns selected without the front-end. Time-to-solution is measured with early stopping on exact recovery, and front-end overhead is measured separately. All reported numbers are produced by the released evaluation scripts from the recorded result artifacts.

Data availability

The external benchmarks (AI-Feynman, PMLB/SRBench) are public[2,8,39]; the physical-law training records are materialised from public formula and range information. The recorded result artifacts that back every figure and table are released with the code.

Code availability

Code for the front-end, training, evaluation and figure generation, together with the result artifacts and the trained graph-attention weights, is released at `https://github.com/chenxi-he/sr-frontend` under the MIT License.

Author contributions

C.H. conceived the study, designed and implemented the front-end and the evaluation pipeline, performed the experiments and analysis, and wrote the manuscript. J.C. supervised the project, advised on the methodology and the physics evaluation, and revised the manuscript. Both authors approved the final version.

Competing interests

The authors declare no competing interests.

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